

Synthesis of β -Amino Alcohols via the Reduction of Lactamides Derived from Ethyl (2*S*)-Lactate with Borane–Methyl Sulfide

Frank W. Lewis,¹ Matthias C. Eichler, David H. Grayson*

Centre for Synthesis and Chemical Biology, University Chemical Laboratory, Trinity College, Dublin 2, Ireland
Fax +353(1671)2826; E-mail: dgrayson@tcd.ie

Received 30 March 2009

Abstract: Reactions of ethyl (2*S*)-lactate with various amines affords lactamides that are reduced with borane–methyl sulfide in the presence of boron trifluoride etherate to generate enantiomerically pure β -amino alcohols in good yield.

Key words: lactamides, borane–methyl sulfide, amino alcohols, asymmetric synthesis, reductions

Chiral β -amino alcohols have widespread applications in asymmetric synthesis, both as ligands for asymmetric catalysis² as well as building blocks in a number of synthetically important chiral auxiliaries.³ The chiral β -amino alcohols **1** and **2** were among the first ligands that were shown to catalyze the highly enantioselective addition of dialkylzinc reagents to aromatic aldehydes,⁴ whilst the Evans' oxazolidinone⁵ **3** and the Meyers' oxazoline⁶ **4** are examples of widely used chiral auxiliaries that are derived from chiral β -amino alcohols. More recently, ligands **5** and **6** gave high enantioselectivities in the asymmetric addition of diphenylzinc to aromatic aldehydes⁷ and the ruthenium-catalyzed transfer hydrogenation of ketones,⁸ respectively (Figure 1).

Amongst the many reported routes to β -amino alcohols,⁹ the most widely employed are the reduction of α -amino acids¹⁰ and the ring opening of epoxides with amines.¹¹ However, a major drawback of the latter route is the cost and availability of the epoxides in optically active form. Enantiopure 2-methyloxirane, for example, presents a significant cost issue, particularly for large-scale syntheses.¹²

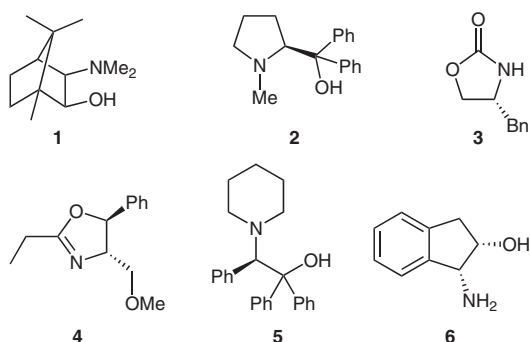
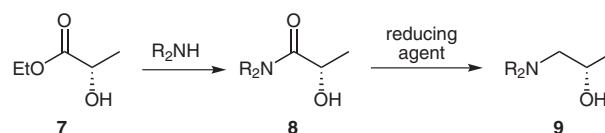


Figure 1

We reasoned that a more cost-effective route to chiral β -amino alcohols of the type that could be obtained by the aminolysis of 2-methyloxirane would involve the reduction of lactamides **8** which can be obtained by either the reactions of amines with ethyl (2*S*)-lactate (**7**), or by the dehydrative condensation of amines with (2*S*)-lactic acid (Scheme 1).¹³ There are a few reports¹⁴ on the reduction of various lactamides **8** but, surprisingly, borane–methyl sulfide does not appear to have been used for this particular transformation. The reduction of analogous α -hydroxyamides derived from mandelic acid and 2-hydroxy-3-methylbutyric acid using borane–methyl sulfide has been reported.¹⁵

Brown and co-workers have shown that the rates of reduction of amides, esters, and nitriles with borane–methyl sulfide can be significantly increased by the distillation of dimethyl sulfide from the reaction mixture and by employing boron trifluoride etherate as Lewis acid.¹⁶ We were thus attracted to the possibility of employing this reagent combination for the synthesis of enantiopure β -amino alcohols **9** from lactamides **8**, as borane–methyl sulfide is more convenient to handle than lithium aluminum hydride. Herein are described our results on the synthesis of various β -amino alcohols **9** derived from ethyl (2*S*)-lactate (**7**).



Scheme 1

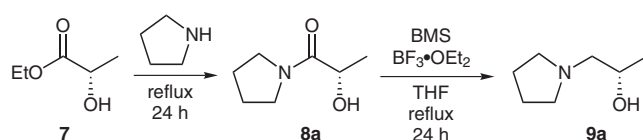
Initially, we focused on the reduction of lactamide **8a** which was obtained¹³ in 82% purified yield by heating ethyl (2*S*)-lactate (**7**) with an excess (1.3 equiv) of pyrrolidine for 24 hours. Treatment of **8a** with borane–methyl sulfide (1 equiv) and boron trifluoride etherate (2 equiv) in THF at reflux for 2.5 hours, followed by aqueous work-up (6 M HCl) according to the procedure of Brown,¹⁶ gave a crude product consisting of the amino alcohol **9a** and unreacted lactamide **8a** in ca. 2:1 ratio as judged by its ¹H NMR spectrum. We considered that the presence of the hydroxy group in **8a** meant that an additional equivalent of borane–methyl sulfide would be needed in order to effect complete reduction. To our satisfaction, we found that the use of 2 equivalents of borane–methyl sulfide and a longer reaction time (24 h) led to complete conversion,

and the desired amino alcohol **9a** was obtained in 65% yield following purification (Scheme 2).

This protocol was subsequently applied to the reduction of a number of lactamides obtained by the reaction of ethyl (2*S*)-lactate (**7**) with various amines,¹⁷ and the results are summarized in Table 1. Our studies focused on aliphatic amines because of the superior performance of aliphatic β -amino alcohols as ligands for the asymmetric addition of organozinc reagents to carbonyl compounds.⁴

The lactamides derived from primary amines (entries 5–10) and from cyclic secondary amines (entries 1–4) were formed in high yields and their β -amino alcohols were obtained in good overall yields for the two steps. However, acyclic secondary amines such as diethylamine failed to react with ethyl (2*S*)-lactate, rendering their derived β -amino alcohols inaccessible by this route. The lactamide derived from allylamine (entry 6) is particularly interesting since reduction of its amide group is accompanied by hydroboration of its double bond. Unfortunately, our attempts to manipulate the intermediate organoborane either by protonolysis (6 M HCl), or by oxidative cleavage (alkaline H₂O₂) of the B–C bond yielded intractable mixtures whose ¹H NMR spectra bore no resemblance to the expected amino alcohol products.

The bislactamides derived from ethylene diamine and 1,6-diaminohexane (entries 11 and 12) each required 4 molar equivalents of borane–methyl sulfide and boron trifluoride etherate for complete reduction to occur. In all other cases (for both secondary and tertiary lactamides) 2 molar equivalents of borane–methyl sulfide and boron trifluoride etherate were used. Notably, lactamides bearing an additional nitrogen atom (entries 13 and 14) did not require an additional equivalent of Lewis acid to effect their complete reduction.



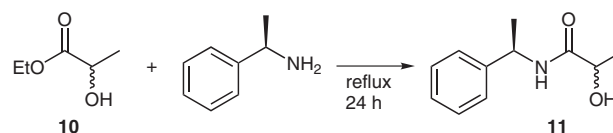
Scheme 2

In a number of the reduction experiments minor amounts of additional products were obtained. These gave rise to a characteristic pair of triplets at $\delta = 3.60$ and 3.69 ppm in the ¹H NMR spectra of the crude products. In many cases, these impurities could not be fully separated from the β -amino alcohol. We suspected that these impurities were oligomers or polymers derived from remaining traces of THF solvent that had undergone acid-catalyzed polymerization during the strongly acidic workup conditions (boiling in 6 M HCl). When a ‘blank’ experiment was run in the absence of lactamide the crude product obtained gave a ¹H NMR spectrum that displayed the same pair of triplets observed in the spectra of the crude reduction products, indicating that these impurities were indeed de-

rived from THF. We then carried out the same reductions in 1,2-dimethoxyethane (DME) as an alternative solvent and were pleased to find that no impurities could be detected in the crude β -amino alcohols, which were obtained in pure form following vacuum distillation or recrystallization.

It is worth noting that in the case of the reductions of lactamides **8k**, **8m**, and **8n** (entries 11, 13, and 14), the crude products were obtained in low overall yields and contained significant quantities of additional impurities (as well as those derived from polymerization of THF) when THF was used as the solvent. These impurities could not be characterized. When DME was used as the solvent, the overall yields were much higher and the crude amino alcohols **9k**, **9m**, and **9n** were obtained free from any impurities. 1,2-Dimethoxyethane is thus the preferred solvent for the reductions of lactamides **8** with borane–methyl sulfide.¹⁸

In order to rule out the possibility that any racemization of the stereogenic centre was occurring during the formation of lactamides by the reaction of amines with ethyl (2*S*)-lactate (**7**), we synthesized lactamides **8i** and **8j** that were derived from each of (1*R*)- α -methylbenzylamine and (1*S*)- α -methylbenzylamine, respectively. The ¹H NMR spectra of both **8i** and **8j** showed that a single diastereomer of the lactamide was present in each case, indicating that the stereochemical integrity of the asymmetric center adjacent to the carbonyl group had been maintained. To fully validate this NMR-based assay we prepared the lactamide **11** from racemic ethyl lactate **10** and (1*R*)- α -methylbenzylamine (Scheme 3). The ¹H NMR spectrum of the product displayed signals due to a pair of diastereomers, of which only one was present in the spectrum of **8i**.



Scheme 3

Thus, we have demonstrated that enantiomerically pure β -amino alcohols can be obtained in a two-step sequence by the reduction of lactamides¹⁹ derived from ethyl (2*S*)-lactate using borane–methyl sulfide/boron trifluoride etherate. These reductions proceeded more cleanly and, in some cases, in higher yield when 1,2-dimethoxyethane was used as the solvent instead of THF. The procedure represents a more cost-effective alternative route to homochiral β -amino alcohols than from syntheses involving the aminolysis of enantiomerically pure 2-methyloxirane. The novel β -amino alcohols that we have prepared are currently under investigation in our laboratory to assess their potential as ligands for the asymmetric addition of diethylzinc to aromatic aldehydes.

Table 1 Synthesis of β -Amino Alcohols **9** from the Reduction of Lactamides **8** Derived from Ethyl (2*S*)-Lactate (**7**) and Various Amines with Borane–Methyl Sulfide

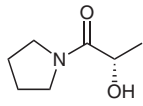
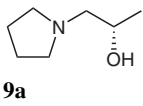
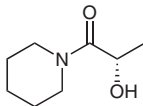
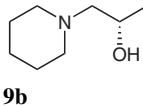
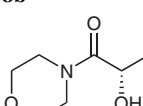
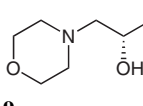
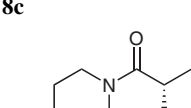
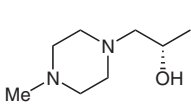
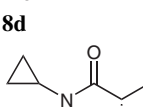
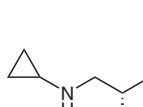
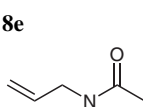
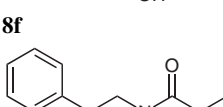
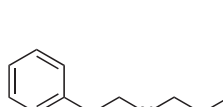
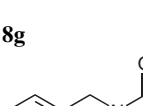
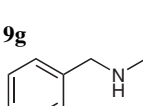
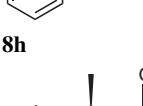
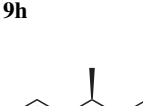
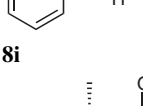
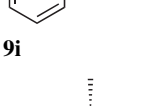
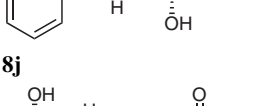
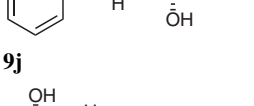
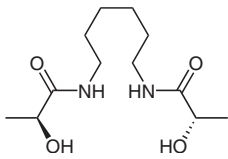
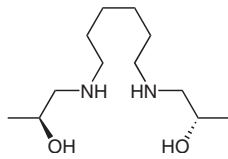
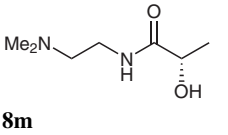
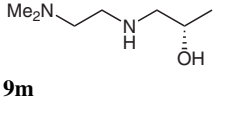
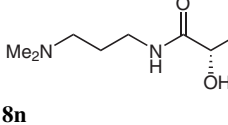
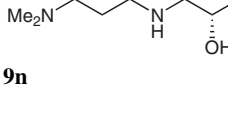
Entry	Amine	Lactamide 8	Yield (%) ^a	β -Amino alcohol 9	Yield (%) ^a
1	pyrrolidine	 8a	82	 9a	65
2	piperidine	 8b	79	 9b	76
3	morpholine	 8c	60	 9c	67
4	<i>N</i> -methylpiperazine	 8d	81	 9d	85
5	cyclopropylamine	 8e	72	 9e	75
6	allylamine	 8f	78	intractable mixture ^b	0
7	2-phenylethylamine	 8g	88	 9g	56
8	benzylamine	 8h	86	 9h	78
9	(1 <i>R</i>)- α -methylbenzylamine	 8i	83	 9i	63
10	(1 <i>S</i>)- α -methylbenzylamine	 8j	81	 9j	65
11	ethane-1,2-diamine	 8k	73	 9k	61

Table 1 Synthesis of β -Amino Alcohols **9** from the Reduction of Lactamides **8** Derived from Ethyl (2*S*)-Lactate (**7**) and Various Amines with Borane–Methyl Sulfide (continued)

Entry	Amine	Lactamide 8	Yield (%) ^a	β -Amino alcohol 9	Yield (%) ^a
12	hexane-1,6-diamine		97		77
13	<i>N,N</i> -dimethylethane-1,2-diamine		97		70
14	<i>N,N</i> -dimethylpropane-1,3-diamine		98		73

^a Yield of lactamide or β -amino alcohol following purification.^b Obtained following either acidic or oxidative workup.

Acknowledgment

The authors gratefully acknowledge Dr. John E. O'Brien and Dr. Manuel Reuther for obtaining NMR spectra and Dr. Martin Feeney for obtaining mass spectra. One of us (F.W.L.) received financial support from Trinity College Dublin.

References and Notes

- (1) New address: F. W. Lewis, Department of Chemistry, University of Reading, Whiteknights, Reading RG6 6AD, UK.
- (2) (a) *Handbook of Reagents for Organic Synthesis: Chiral Reagents for Asymmetric Synthesis*; Paquette, L. A., Ed.; Wiley: Chichester, **2003**. (b) *Asymmetric Synthesis – The Essentials*; Christmann, M.; Bräse, S., Eds.; Wiley-VCH: Weinheim, **2007**.
- (3) (a) *Compendium of Chiral Auxiliary Applications*; Roos, G., Ed.; Academic Press: New York, **2002**. (b) Gnäs, Y.; Glorius, F. *Synthesis* **2006**, 1899.
- (4) (a) Soai, K.; Niwa, S. *Chem. Rev.* **1992**, 92, 833. (b) Knochel, P.; Singer, R. D. *Chem. Rev.* **1993**, 93, 2117. (c) Pu, L.; Yu, H.-B. *Chem. Rev.* **2001**, 101, 757.
- (5) (a) Evans, D. A. *Aldrichimica Acta* **1982**, 15, 23. (b) Ager, D. J.; Prakash, I.; Schaad, D. R. *Aldrichimica Acta* **1997**, 30, 3.
- (6) (a) Meyers, A. I.; Lutomski, K. A. *Acc. Chem. Res.* **1978**, 11, 375. (b) Gant, T. G.; Meyers, A. I. *Tetrahedron* **1994**, 50, 2297. (c) Meyers, A. I. *J. Org. Chem.* **2005**, 70, 6137.
- (7) Dimitrov, V.; Kostova, K. *Lett. Org. Chem.* **2006**, 3, 176.
- (8) Gladiali, S.; Alberico, E. *Chem. Soc. Rev.* **2006**, 35, 226.
- (9) From the reduction of α -amino ketones: (a) Tramontini, M. *Synthesis* **1982**, 605. (b) Hoffman, R. V.; Maslouh, N.; Cervantes-Lee, F. J. *J. Org. Chem.* **2002**, 67, 1045. (c) Fraser, D. S.; Park, S. B.; Chong, J. M. *Can. J. Chem.* **2004**, 82, 87. From the manipulation of cyanohydrins: (d) Jackson, W. R.; Jacobs, H. A.; Jayatilake, G. S.; Matthews, B. R.; Watson, K. G. *Aust. J. Chem.* **1990**, 43, 2045. From nucleophilic additions to α -amino aldehydes: (e) Reetz, M. T. *Angew. Chem., Int. Ed. Engl.* **1991**, 30, 1531. From nucleophilic additions to chiral imine derivatives: (f) Tang, T. P.; Volkman, S. K.; Ellman, J. A. *J. Org. Chem.* **2001**, 66, 8772. (g) Barrow, J. C.; Ngo, P. L.; Pellicore, J. M.; Selnick, H. G.; Nantermet, P. G. *Tetrahedron Lett.* **2001**, 42, 2051. From the reduction of nitrogen-stabilized oxallyl cation [4+3] cycloadducts: (h) Huang, J.; Ianni, J. C.; Antoline, J. E.; Hsung, R. P.; Kozlowski, M. C. *Org. Lett.* **2006**, 8, 1565.
- (10) (a) Karrer, P.; Portmann, P.; Suter, M. *Helv. Chim. Acta* **1948**, 31, 1617. (b) Seki, H.; Koga, K.; Matsuo, H.; Ohki, S.; Matsuo, I.; Yamada, S. *Chem. Pharm. Bull.* **1965**, 13, 995. (c) Prasad, B.; Saund, A. K.; Bora, J. M.; Mathur, N. K. *Ind. J. Chem.* **1974**, 12, 290. (d) Smith, G. A.; Gawley, R. E. *Org. Synth.* **1985**, 63, 136. (e) Gage, J. R.; Evans, D. A. *Org. Synth.* **1989**, 68, 77. (f) Giannis, A.; Sandhoff, K. *Angew. Chem., Int. Ed. Engl.* **1989**, 28, 218. (g) Dickman, D. A.; Meyers, A. I.; Smith, G. A.; Gawley, R. E. *Organic Syntheses*, Collect. Vol. VII; Wiley and Sons: New York, **1990**, 530. (h) Dharanipragada, R.; Alarcon, A.; Hruby, V. J. *Org. Prep. Proced. Int.* **1991**, 23, 396. (i) Abiko, A.; Masamune, S. *Tetrahedron Lett.* **1992**, 33, 5517. (j) McKennon, M. J.; Meyers, A. I.; Drauz, K.; Schwarm, M. *J. Org. Chem.* **1993**, 58, 3568.
- (11) (a) Hodgson, D. M.; Gibbs, A. R.; Lee, G. P. *Tetrahedron* **1996**, 52, 14361. (b) *Comprehensive Organic Synthesis*, Part 1.3.4.1., Vol. 6; Mitsunobu, O.; Winterfeldt, E., Eds.; Pergamon: New York, **1996**. (c) Ager, D. J.; Prakash, I.; Schaad, D. R. *Chem. Rev.* **1996**, 96, 835. (d) Bergmeier, S. C. *Tetrahedron* **2000**, 56, 2561.
- (12) Cost per gram from Aldrich Chemical Company: ethyl (2*S*)-lactate: £0.05; (2*S*)-2-methyloxirane: £8.66.
- (13) (a) Ratchford, W. P.; Fisher, C. H. *J. Org. Chem.* **1950**, 15, 317. (b) Fein, M. L.; Filachione, E. M. *J. Am. Chem. Soc.* **1953**, 75, 2097. (c) Shapiro, S. L.; Rose, I. M.; Freedman, L. *J. Am. Chem. Soc.* **1959**, 81, 6322. (d) Dukes, M.; Smith, L. H. *J. Med. Chem.* **1971**, 14, 326.
- (14) With LiAlH_4 : (a) See ref. 13d. (b) Yoneyama, K.; Ichizen, N.; Konnai, M.; Takematsu, T.; Ushinohama, K.; Jikihara, T. *Agric. Biol. Chem.* **1984**, 48, 995. (c) Lai, J.-Y.; Shi, X.-X.; Gong, Y.-S.; Dai, L.-X. *J. Org. Chem.* **1993**, 58, 4775. With $\text{NaBH}_4/\text{H}_2\text{SO}_4$: (d) Taoka, N.; Yasohara, Y.; Yao, T.; Maehara, K.; Ueda, Y. WO 2002070482, **2002**.
- (15) Cobb, A. J. A.; Marson, C. M. *Tetrahedron* **2005**, 61, 1269.

- (16) (a) Brown, H. C.; Narasimhan, S.; Choi, Y. M. *Synthesis* **1981**, 996. (b) Brown, H. C.; Choi, Y. M.; Narasimhan, S. *J. Org. Chem.* **1982**, 47, 3153.

(17) **Synthesis of Lactamides 8a–n – General Procedure**

Ethyl (2*S*)-lactate (**7**, 88.2 mmol, 1 equiv) and the appropriate amine (114.6 mmol, 1.3 equiv) were placed in a flask and heated under reflux for 24 h. The solution was allowed to cool to r.t., the volatiles were removed under reduced pressure ($1.33 \cdot 10^{-4}$ bar), and the lactamide was purified by vacuum distillation or recrystallization. For the synthesis of bislactamides **8k** and **8l**, 0.5 equiv of amine were used. Compounds **8a–d,f–j,l–n** are known.²⁰

Characterization Data for 8e

Yellow oil, bp 101–104 °C ($1.33 \cdot 10^{-4}$ bar). $[\alpha]_D^{16}$ –16.6 (*c* 1.31, CHCl₃). IR (neat): 3308 (OH, NH), 2978, 1655 (C=O), 1532, 1457, 1364, 1275, 1124, 1048, 978 cm⁻¹. ¹H NMR (400.1 MHz, CDCl₃): δ = 6.90 (s, 1 H, exch. D₂O, NH), 4.18 (q, *J* = 6.5 Hz, 1 H, CHOH), 3.84 (br s, 1 H, exch. D₂O, OH), 2.67–2.74 (m, 1 H, CH), 1.39 (d, *J* = 6.5 Hz, 3 H, CH₃), 0.72–0.84 (m, 2 H, CH₂), 0.46–0.58 (m, 2 H, CH₂). ¹³C NMR (100.6 MHz, CDCl₃): δ = 176.1 (C=O), 67.7 (CHOH), 21.6 (CH), 20.6 (CH₃), 5.9 (CH₂), 5.8 (CH₂). HRMS (EI, MeOH): *m/z* calcd for C₆H₁₁NO₂ [*M* + Na]⁺: 152.0687; found: 152.0680.

Characterization Data for 8k

White solid, mp 97–98 °C (10% EtOH in THF). $[\alpha]_D^{16}$ –24.7 (*c* 1.96, MeOH). IR (Nujol): 3323 (OH), 3086 (NH), 2926, 1662 (C=O), 1561, 1458, 1368, 1310, 1274, 1223, 1129, 1079, 937 cm⁻¹. ¹H NMR (400.1 MHz, DMSO-*d*₆): δ = 7.81 (s, 1 H, exch. D₂O, NH), 5.48 (s, 1 H, exch. D₂O, NH), 3.94 (q, *J* = 6.5 Hz, 2 H, 2 $\times$ CHOH), 3.36 (br s, 2 H, exch. D₂O, 2 $\times$ OH), 3.16 (app t, *J* = 2.5 Hz, 4 H, 2 $\times$ CH₂NH), 1.19 (d, *J* = 6.5 Hz, 6 H, 2 $\times$ CH₃). ¹³C NMR (100.6 MHz, DMSO-*d*₆): δ = 174.8 (2 $\times$ C=O), 67.2 (2 $\times$ CHOH), 38.2 (2 $\times$ CH₂NH), 21.0 (2 $\times$ CH₃). HRMS (EI, MeOH): *m/z* calcd for C₈H₁₆N₂O₄ [*M* + Na]⁺: 227.1008; found: 227.1010.

(18) **Synthesis of β -Amino Alcohols 9a–n – General Procedure**

The lactamide **8** (23.7 mmol, 1 equiv) was dissolved in dry 1,2-dimethoxyethane (15 mL/g) in an oven-dried flask equipped with a Dean–Stark trap and reflux condenser under a nitrogen atmosphere. Boron trifluoride etherate (47.5 mmol, 2 equiv) was added, and then borane–methyl sulfide (47.5 mmol, 2 equiv) was added dropwise via syringe. Once the evolution of H₂ had ceased, the solution was heated under reflux for 24 h (ca. 2–3 mL of distillate collected) and the solvents were removed in vacuo. The resulting colorless oil was carefully diluted with 6 M HCl (15 mL/g) and boiled for ca. 5 min. The resulting clear solution was neutralized while still hot with 6 M NaOH (15 mL/g) and then allowed to cool to r.t. before being saturated with solid K₂CO₃ and extracted with CHCl₃ (3 $\times$ 50 mL). The combined organic extracts were dried over MgSO₄ and evaporated to afford the crude β -amino alcohol **9** which was purified by vacuum distillation or recrystallization. In the case of bislactamides **8k** and **8l**, 4 equiv of both boron trifluoride etherate and borane–methyl sulfide were used. Compounds **9a–d,f–k** are known.²¹

Characterization Data for 9e

Colorless liquid, bp 76 °C ($1.33 \cdot 10^{-2}$ to $2.66 \cdot 10^{-2}$ bar). $[\alpha]_D^{16}$ +41.9 (*c* 4.95, CHCl₃). IR (neat): 3307 (OH, NH), 2965, 2931, 1451, 1373, 1141, 1103, 1066, 1015, 917, 816 cm⁻¹. ¹H NMR (400.1 MHz, CDCl₃): δ = 3.79 (dq, *J* = 10.0, 6.5, 3.0 Hz, 1 H, CHOH), 2.83 (dd, *J* = 12.0, 3.0 Hz, 1 H, CH₂NH), 2.50 (dd, *J* = 12.0, 10.0 Hz, 1 H, CH₂NH), 2.28 (br s, 2 H, exch. D₂O, OH and NH), 2.13–2.18 (m, 1 H, CH), 1.16 (d, *J* = 6.5 Hz, 3 H, CH₃), 0.41–0.51 (m, 2 H, CH₂), 0.29–0.40 (m, 2 H, CH₂). ¹³C NMR (100.6 MHz, CDCl₃):

δ = 65.0 (CHOH), 56.4 (CH₂NH), 29.8 (CH), 19.9 (CH₃), 6.5 (CH₂), 5.4 (CH₂). Anal. Calcd for C₆H₁₃NO: C, 62.61; H, 11.30; N, 12.17. Found: C, 62.39; H, 11.58; N, 11.95.

Characterization Data for 9l

White solid, mp 83–84 °C (CHCl₃). $[\alpha]_D^{17}$ +34.2 (*c* 3.99, CHCl₃). IR (Nujol): 3273 (OH), 3181 (NH), 2914, 1456, 1366, 1135, 1084, 985, 935, 900, 846 cm⁻¹. ¹H NMR (400.1 MHz, CDCl₃): δ = 3.80 (dq, *J* = 9.5, 6.0, 3.0 Hz, 2 H, 2 $\times$ CHOH), 2.93 (br s, 4 H, exch. D₂O, 2 $\times$ NH and 2 $\times$ OH), 2.70 (dd, *J* = 12.0, 3.0 Hz, 2 H, CH₂CHOH), 2.53–2.70 (m, 4 H, 2 $\times$ CH₂NH), 2.42 (dd, *J* = 12.0, 9.5 Hz, 2 H, CH₂CHOH), 1.46–1.57 (m, 4 H, 2 $\times$ CH₂CH₂NH), 1.31–1.39 (m, 4 H, 2 $\times$ CH₂CH₂CH₂NH), 1.16 (d, *J* = 6.0 Hz, 6 H, 2 $\times$ CH₃). ¹³C NMR (100.6 MHz, CDCl₃): δ = 65.3 (2 $\times$ CHOH), 56.8 (2 $\times$ CH₂NH), 49.4 (2 $\times$ CH₂CHOH), 29.8 (2 $\times$ CH₂CH₂NH), 27.0 (2 $\times$ CH₂CH₂CH₂NH), 20.7 (2 $\times$ CH₃). HRMS (EI, MeOH): *m/z* calcd for C₁₂H₂₈N₂O₂ [*M* + H]⁺: 233.2229; found: 233.2234.

Characterization Data for 9m

Colorless liquid, bp 62 °C ($1.33 \cdot 10^{-4}$ bar). $[\alpha]_D^{16}$ +34.2 (*c* 3.61, CHCl₃). IR (neat): 3304 (OH, NH), 2942, 2818, 1460, 1370, 1277, 1129, 1055, 846, 778 cm⁻¹. ¹H NMR (400.1 MHz, CDCl₃): δ = 3.78 (dq, *J* = 9.5, 6.0, 3.0 Hz, 1 H, CHOH), 2.65–2.78 (m, 3 H, CH₂CHOH and CH₂NH), 2.42 (dd, *J* = 12.0, 9.5 Hz, 1 H, CH₂CHOH), 2.37–2.46 [m, 2 H, CH₂N(CH₃)₂], 2.23 [s, 6 H, N(CH₃)₂], 1.15 (d, *J* = 6.0 Hz, 3 H, CH₃). ¹³C NMR (100.6 MHz, CDCl₃): δ = 65.1 (CHOH), 58.7 [CH₂N(CH₃)₂], 56.5 (CH₂CHOH), 46.3 (CH₂NH), 45.0 [N(CH₃)₂], 20.0 (CH₃). HRMS (EI, MeOH): *m/z* calcd for C₇H₁₈N₂O [*M* + H]⁺: 146.1181; found: 146.1187.

Characterization Data for 9n

Colorless liquid, bp 72–76 °C ($1.33 \cdot 10^{-4}$ bar). $[\alpha]_D^{16}$ +37.2 (*c* 1.68, CHCl₃). IR (neat): 3303 (OH, NH), 2941, 2816, 1461, 1372, 1265, 1127, 1061, 839 cm⁻¹. ¹H NMR (400.1 MHz, CDCl₃): δ = 3.79 (dq, *J* = 9.5, 6.3, 3.0 Hz, 1 H, CHOH), 2.70 (dd, *J* = 12.1, 3.0 Hz, 1 H, CH₂CHOH), 2.61–2.75 (m, 2 H, CH₂NH), 2.41 (dd, *J* = 12.1, 9.5 Hz, 1 H, CH₂CHOH), 2.34 [t, *J* = 7.2 Hz, 2 H, CH₂N(CH₃)₂], 2.23 [s, 6 H, N(CH₃)₂], 1.66 (q, *J* = 7.0 Hz, 2 H, CH₂), 1.15 (d, *J* = 6.3 Hz, 3 H, CH₃). ¹³C NMR (100.6 MHz, CDCl₃): δ = 65.3 (CHOH), 57.7 [CH₂N(CH₃)₂], 56.7 (CH₂CHOH), 47.6 (CH₂NH), 45.5 [N(CH₃)₂], 27.9 (CH₂), 20.4 (CH₃). HRMS (EI, MeOH): *m/z* calcd for C₈H₂₀N₂O [*M* + H]⁺: 161.1654; found: 161.1646.

- (19) Lactamides **8a–d** have recently been used as chiral auxiliaries, see: Ammazalorso, A.; Amoroso, R.; Bettoni, G.; De Filippis, B.; Fantacuzzi, M.; Giampietro, L.; Maccallini, C.; Tricca, M. L. *Eur. J. Org. Chem.* **2006**, 4088.
- (20) Compound **8a**: (a) See ref. 13a. (b) Ito, Y.; Kobayashi, Y.; Kawabata, T.; Takase, M.; Terashima, S. *Tetrahedron* **1989**, 45, 5767. (c) Schurig, V.; Betschinger, F. *Bull. Soc. Chim. Fr.* **1994**, 131, 555. Compound **8b**: (d) See ref. 13a. (e) Schurig, V.; Hintzer, K.; Leyrer, U.; Mark, C.; Pitchen, P.; Kagan, H. B. *J. Organomet. Chem.* **1989**, 370, 81. (f) See ref. 20c. Compound **8c**: (g) See ref. 13a. (h) Tasaka, A.; Tamura, N.; Matsushita, Y.; Teranishi, K.; Hayashi, R. *Chem. Pharm. Bull.* **1993**, 41, 1035. (i) Upadhayaya, R.-S.; Sinha, N.; Jain, S.; Kishore, N.; Chandra, R.; Arora, S.-K. *Bioorg. Med. Chem.* **2004**, 12, 2225. Compound **8d**: (j) See ref. 19. Racemic **8f**: (k) See ref. 13a.c. Compound **8g**: (l) Adamczyk, M.; Grote, J.; Rege, S. *Tetrahedron: Asymmetry* **1997**, 8, 2509. Compound **8h**: (m) Savinov, S. N.; Austin, D. J. *Org. Lett.* **2002**, 4, 1415. Compound **8i**: (n) See ref. 14c. Compound **8j**: (o) Harada, K.; Munegumi, T.; Nomoto, S. *Tetrahedron Lett.* **1981**, 22, 111. (p) Harada, K.; Munegumi, T. *Bull. Chem. Soc. Jpn.* **1984**, 57, 3203. (q) See ref. 14c. Compound **8l**: (r) Kern, T.

- Makromol. Chem.* **1955**, *16*, 89. Compound **8m**: (s) Tounsi, N.; Dupont, L.; Mohamadou, A.; Cadiou, C.; Aplingcourt, M.; Plantier-Royon, R.; Massicot, F.; Portella, C. *New J. Chem.* **2004**, *28*, 785. Compound **8n**: (t) See ref. 20s.
- (21) Racemic **9a**: (a) Reid, W. B.; Wright, J. B. Jr.; Kolloff, H. G.; Hunter, J. H. *J. Am. Chem. Soc.* **1948**, *70*, 3100. (b) Halverstadt, I. F.; Hardie, W. R.; Williams, A. R. *J. Am. Chem. Soc.* **1959**, *81*, 3618. (c) Azizi, N.; Saidi, M. R. *Org. Lett.* **2005**, *7*, 3649. Compound **9b**: (d) Rojkovicova, T.; Lehotay, J.; Cizmarik, J. *Pharmazie* **2003**, *58*, 483. (e) Martin-Matute, B.; Edin, M.; Bogar, K.; Kaynak, F. B.; Baekvall, J.-E. *J. Am. Chem. Soc.* **2005**, *127*, 8817. Compound **9c**: (f) Angeloni, A. S. *Gazz. Chim. Ital.* **1977**, *107*, 421. Racemic **9d**: (g) Redel, B. *Bull. Soc. Chim. Fr.* **1955**, *92*, 1411. (h) Bourquin, J.-P.; Schwarb, G.; Gamboni, G.; Fischer, R.; Ruesch, L.; Guldemann, S.; Theus, V.; Schenker, E.; Renz, J. *Helv. Chim. Acta* **1958**, *41*, 1072. Racemic **9g**: (i) Beckett, M. *J. Pharm. Pharmacol.* **1968**, *20*, 218. Compound **9h**: (j) Kashima, C.; Harada, K. *J. Chem. Soc., Perkin Trans. 1* **1988**, 1521. (k) Brun, E. M.; Gil, S.; Parra, M. *Tetrahedron: Asymmetry* **2001**, *12*, 915. (l) Gawronski, J.; Kolbon, H.; Kwit, M. *Enantiomer* **2002**, *7*, 85. Compound **9i**: (m) See ref. 14b. (n) See ref. 14c. Compound **9j**: (o) See ref. 14b. (p) See ref. 14c. Racemic **9k**: (q) Steck, E. A.; Buck, J. S.; Fletcher, L. T. *J. Am. Chem. Soc.* **1957**, *79*, 4414. (r) Hall, J. L.; Dean, W. E.; Pacovsky, E. A. *J. Am. Chem. Soc.* **1960**, *82*, 3303. (s) Butula, I.; Karlovic, G. *Justus Liebigs Ann. Chem.* **1976**, 1455.

Copyright of Synlett is the property of Georg Thieme Verlag Stuttgart and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.